

STERLITAMAK, Russian Federation

WMO: 288250

Lat: 53.6036N

Lon: 55.9329E

Elev: 450

StdP: 14.46

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.8	-17.0	-28.7	1.1	-22.7	-22.5	1.6	-16.4	14.3	21.3	12.8	23.2	0.9	0	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	89.9	66.8	86.7	66.1	83.5	64.9	69.8	83.9	68.3	81.9	66.9	79.4	4.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	94.9	75.2	63.7	89.9	74.0	62.2	85.0	72.7	34.1	83.7	32.8	81.9	31.7	79.9	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.3	9.8	DB	-28.9	94.3	6.1	2.6	-33.2	96.2	-36.8	97.7	-40.2	99.2	-44.6	101.1	(4)
(5)			WB	-29.1	72.1	6.0	1.8	-33.4	73.4	-37.0	74.4	-40.4	75.4	-44.8	76.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.2	8.7	11.4	23.6	43.2	57.9	65.3	68.8	65.8	54.8	41.4	26.6	13.7		
	DBStd	23.89	13.47	12.94	10.23	10.19	8.31	7.81	6.45	7.76	7.70	8.24	11.34	13.90		
	HDD50	5616	1282	1082	818	253	26	2	0	1	37	287	702	1126		
	HDD65	9440	1747	1502	1283	655	247	92	37	86	316	732	1152	1591		
	CDD50	2052	0	0	0	49	271	460	581	490	181	20	0	0		
	CDD65	403	0	0	0	1	27	100	154	111	10	0	0	0		
	CDH74	4371	0	0	0	24	434	1130	1511	1140	132	0	0	0		
	CDH80	1535	0	0	0	2	111	416	524	448	34	0	0	0		
	WSAvg	3.9	3.9	4.3	4.4	4.3	4.2	3.9	3.6	3.3	3.1	3.9	4.2	4.2		
	PrecAvg	21.6	1.7	1.4	1.5	1.2	1.8	2.2	2.1	2.2	1.6	2.0	1.9	2.0		
	PrecMax	30.2	3.4	3.2	3.5	3.5	6.2	4.5	6.3	6.0	3.0	4.3	3.7	4.8		
(15) Precipitation	PrecMin	13.8	0.3	0.0	0.1	0.0	0.2	0.1	0.1	0.4	0.4	0.7	0.2	0.2		
	PrecStd	4.1	0.9	0.7	0.9	0.8	1.3	1.3	1.3	1.1	0.8	1.0	1.1	0.9		
	DB	34.7	37.3	51.2	77.4	87.0	92.7	93.1	94.2	84.8	69.5	51.4	36.5			
	MCWB	32.9	32.9	41.6	56.4	62.0	67.0	68.7	66.8	61.4	52.8	46.9	33.8			
	2% DB	32.4	34.2	43.2	71.2	83.3	88.9	88.8	89.4	77.4	63.9	45.6	34.6			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	30.6	32.1	36.8	54.0	60.6	66.1	68.3	66.2	59.7	50.3	42.6	32.8			
	5% DB	30.2	32.1	39.3	66.0	79.1	85.0	85.6	85.3	72.5	58.7	42.2	33.2			
	MCWB	28.5	30.3	34.6	51.4	59.7	65.4	67.1	66.1	58.2	48.4	39.2	31.6			
	10% DB	27.5	29.1	36.5	60.0	74.5	80.8	82.5	80.5	67.8	53.7	38.8	31.4			
	MCWB	25.7	27.1	33.1	48.5	57.2	63.7	65.9	64.5	56.1	46.9	36.3	29.6			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	33.3	34.2	42.0	57.9	65.4	70.7	72.2	70.9	64.5	55.6	47.9	35.0			
	MCDB	34.4	35.8	50.0	74.0	79.1	85.5	86.4	84.9	79.6	64.5	51.7	35.9			
	2% WB	30.9	32.4	37.9	55.0	63.0	68.6	70.3	68.8	61.8	52.5	43.1	33.2			
	MCDB	32.2	33.9	42.1	68.2	77.0	83.3	84.0	83.6	72.9	60.6	45.0	34.2			
	5% WB	28.5	30.3	35.4	52.3	61.1	66.9	68.9	67.2	59.7	49.8	39.3	31.8			
(33) Mean Daily Temperature Range	MCDB	30.1	32.1	38.2	64.3	75.6	80.6	81.8	81.1	69.3	56.8	41.5	33.2			
	10% WB	25.8	27.2	33.6	49.0	59.1	65.2	67.5	65.6	57.6	47.4	36.8	29.6			
	MCDB	27.4	29.1	36.3	59.4	72.2	77.8	79.1	78.1	65.9	53.0	38.6	31.2			
	5% DB	13.1	15.8	16.1	17.7	21.5	20.3	19.4	19.8	18.3	13.7	9.3	11.9			
	MCDBR	9.7	11.9	15.8	26.5	29.4	27.4	25.1	27.2	27.0	25.4	10.1	9.2			
(40) Clear-Sky Solar Irradiance	MCWBR	9.2	10.5	11.1	12.4	10.9	9.5	8.2	8.7	11.5	13.8	8.0	8.6			
	5% WB	9.3	10.3	12.5	24.7	26.2	23.2	21.6	23.7	22.6	20.8	9.3	9.4			
	MCDBR	8.9	9.5	9.4	12.2	10.9	9.3	7.9	9.1	11.0	12.4	8.1	9.1			
	taub	0.299	0.325	0.377	0.423	0.406	0.399	0.413	0.435	0.384	0.355	0.326	0.300			
	taud	2.046	2.013	1.939	2.057	2.251	2.329	2.291	2.209	2.321	2.298	2.172	2.027			
(42) All-Sky Solar Radiation	Ebn at Noon	197	234	246	253	267	270	263	248	248	226	189	168			
	Edh at Noon	23	34	46	46	40	38	39	40	31	26	21	20			
	RadAvg	307	639	1087	1425	1714	1880	1841	1476	971	466	219	197			
(45)	RadStd	36	59	108	145	161	217	165	149	117	98	48	39			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page